**V1 frequency dispersion and excitation on a square torus**

Explicit spectral derivations and interpretation of the simulations

This report extends the neural-field analysis in `final_report.pdf`. All new simulations use the three- and four-population calibrations specified in Report 1. We distinguish exact statements, a local interval certificate, ordinary numerical calculations, and a formal weak-forcing expansion.

# 1 Field equation and the dispersion matrix

Let $u : \mathbb{T}^d_L \times \mathbb{R}_+ \to \mathbb{R}^N$, with $N = 3, 4$ and $d = 1, 2$, satisfy

$$D\partial_t u = -u + W * f(u) + I(x,t), \qquad D = \mathrm{diag}(\tau_1, \dots, \tau_N), \quad f(0) = 0. \tag{1}$$

The convolution applies the scalar nonlinearity to each *source* population before integration. Rows index targets and columns index sources. Our kernels are

$$W_{ij}(x;p) = J_{ij}(p) K_{\Sigma_{ij}}(x), \quad K_\sigma(x) = \sum_{r\in\mathbb{Z}^d} \frac{\exp(-|x+Lr|^2/(2\sigma))}{(2\pi\sigma)^{d/2}}, \quad J(p) = (0.5+p)G. \tag{2}$$

Each $K_\sigma$ has integral one over the torus. With $\widehat{u}_k = L^{-d}\int_{[0,L]^d} u(x)\mathrm{e}^{-\mathrm{i}q_k\cdot x}dx$, $q_k = 2\pi k/L$, convolution has multiplier

$$\widehat{W}(q;p) = J(p) \odot \exp(-\Sigma|q|^2/2). \tag{3}$$

Here $\widehat{W}$ denotes the integral multiplier, not the normalized Fourier coefficient of $W$, and $\odot$ and the exponential act entrywise. Writing $S = \mathrm{diag}(s_j)$, $s_j = f_j'(0)$, and substituting $u = v\mathrm{e}^{\lambda t + \mathrm{i}q_k\cdot x}$ into the linearization at $I = 0, u = 0$ gives

$$\lambda D v = -v + \widehat{W}(q_k;p) S v, \qquad \boxed{M(q,p) = D^{-1}[-I_N + \widehat{W}(q;p)S]}, \quad q = |q_k|. \tag{4}$$

Thus $P(\lambda, q, p) = \det(\lambda I_N - M(q,p)) = 0$ is the dispersion relation. For the sigmoid in Report 1, $s_j = g_j \ell(-g_j\theta_j)[1 - \ell(-g_j\theta_j)]$. Time is in ms; space is in 0.1 mm units. Therefore

$$q_{\mathrm{phys}} = 10q \text{ rad/mm}, \qquad f_{\mathrm{Hz}} = \frac{1000}{2\pi}\omega, \qquad \omega = \mathrm{Im}\,\lambda. \tag{5}$$

At the calibrated onset, $p_0 = 0.25$, $k_0 = 3$, $q_0 = 6\pi/47.6023 = 0.3959799405$ and $\omega_0 = 0.08$ rad/ms.

## 2 What is the imaginary part of a dominant eigenvalue?

Define $a(q) = \max_{\lambda \in \operatorname{spec} M(q)} \operatorname{Re} \lambda$. When a unique real root, or one conjugate pair, has this real part, define $\omega_d(q) = 0$ or the positive imaginary part of that pair, respectively. At a tie between unrelated roots, there is no unique physical frequency. For a single plotted curve we use

$$\omega_d(q) = \max\{\operatorname{Im} \lambda : \lambda \in \operatorname{spec} M(q),\ \operatorname{Re} \lambda = a(q)\}. \tag{6}$$

This convention can be discontinuous. It is different from continuing the same complex eigenvalue after it ceases to dominate.

**Proposition 1** (Exact frequency derivative)**.** *On an interval where $\lambda(q) = \alpha(q) + \mathrm{i}\omega(q)$ is a simple eigenvalue, let $Mv = \lambda v$ and $w^* M = \lambda w^*$, normalized by $w^* v = 1$. Then*

$$\lambda' = w^* M' v = -\frac{P_q}{P_\lambda}, \qquad M' = -qD^{-1}[(\Sigma \odot J \odot \mathrm{e}^{-\Sigma q^2/2})S]. \tag{7}$$

*If $P_\lambda = A + \mathrm{i}B$ and $P_q = C + \mathrm{i}E$, evaluated at the root, then*

$$\boxed{\omega'(q) = \frac{CB - EA}{A^2 + B^2}}. \tag{8}$$

*Consequently a simple dominant pair increases strictly on any interval where $CB - EA > 0$ throughout and dominance does not change.*

*Proof.* Differentiate $Mv = \lambda v$ and multiply by $w^*$. The two $v'$ terms cancel. Alternatively differentiate $P(\lambda(q), q) = 0$; simplicity gives $P_\lambda \neq 0$. Multiplying $-P_q/P_\lambda$ by $\overline{P_\lambda}/\overline{P_\lambda}$ gives (8). Positive derivative implies strict increase by integration. □

No sign follows merely from Gaussian attenuation. If $R_{ij} = J_{ij} s_j \mathrm{e}^{-\Sigma_{ij} q^2/2}/\tau_i$, then

$$\omega' = -q \sum_{ij} \Sigma_{ij} R_{ij} \operatorname{Im}(\overline{w}_i v_j). \tag{9}$$

Both the signed edges and the complex eigenvectors matter. Even if all absolute couplings decrease, the imaginary part of their collective eigenvalue can grow.

For comparison, if all $\tau_i = \tau$ and all retained widths equal $\sigma$, then $M = (-I + hJS)/\tau$, $h = \mathrm{e}^{-\sigma q^2/2}$. For every eigenvalue $\nu$ of $JS$,

$$\lambda_\nu(q) = (-1 + h\nu)/\tau, \qquad \omega'_\nu = -\sigma q \omega_\nu < 0 \quad (q > 0,\ \omega_\nu > 0). \tag{10}$$

Thus connectivity topology alone cannot prove increasing frequency. Unequal widths and time scales require the actual sign calculation in (8).

## 3 Explicit characteristic polynomials for the V1 motif

At each $(q, p)$ write the two matrices as

$$M_3 = \begin{pmatrix} a & b & c \\ d & e & f \\ g & 0 & h \end{pmatrix}, \qquad M_4 = \begin{pmatrix} a & b & c & 0 \\ d & e & f & 0 \\ g & 0 & h & j \\ r & 0 & s & t \end{pmatrix}. \tag{11}$$

Every entry is explicit from (4); in particular

$$(M_{ij})' = -q\Sigma_{ij}(M_{ij} + \delta_{ij}/\tau_i). \tag{12}$$

Define $H = (\lambda - a)(\lambda - e) - bd$ and $R = c(\lambda - e) + bf$. Expanding the $3 \times 3$ determinant yields

$$P_3 = (\lambda - h)H - gR = \lambda^3 + A\lambda^2 + B\lambda + C, \tag{13}$$

where

$$\begin{aligned} A &= -a - e - h, \\ B &= ae - bd + h(a + e) - gc, \\ C &= -h(ae - bd) + gce - gbf. \end{aligned} \tag{14}$$

For four populations, eliminate the last component when $\lambda \neq t$: the third row acquires $jr/(\lambda - t)$ and $js/(\lambda - t)$ in its first and third entries. Multiplying the reduced determinant by $\lambda - t$ gives

$$\boxed{P_4 = (\lambda - t)P_3 - j(sH + rR)}. \tag{15}$$

Both sides are polynomials, so equality also holds at $\lambda = t$. The $P_3$ in this identity is the four-population matrix's upper block; it is not the independently recalibrated three-population model. Writing $P_4 = \lambda^4 + c_1\lambda^3 + c_2\lambda^2 + c_3\lambda + c_4$ gives

$$\begin{aligned} c_1 &= A - t, & c_2 &= B - tA - js, \\ c_3 &= C - tB + js(a + e) - jrc, \\ c_4 &= -tC - js(ae - bd) + jrce - jrbf. \end{aligned} \tag{16}$$

Equations (12)–(16), differentiated by the product rule, provide every coefficient and its $q$ derivative directly from $G, \Sigma, D, S$. No fitted polynomial or eigenvalue finite difference is needed.

For three populations with one real root $\rho$ and a complex pair, factorization gives

$$\begin{aligned} \alpha &= -\tfrac{1}{2}(A + \rho), & \omega^2 &= B - 2\alpha\rho - \alpha^2, \\ \rho' &= -\frac{A'\rho^2 + B'\rho + C'}{3\rho^2 + 2A\rho + B}, & \alpha' &= -\tfrac{1}{2}(A' + \rho'), \\ \omega' &= \frac{B' - 2(\rho + \alpha)\alpha' - 2\alpha\rho'}{2\omega}. \end{aligned} \tag{17}$$

The root $\rho$ is the unique real zero of (13) in this case; $\omega^2 > 0$ selects a pair and $\alpha > \rho$ checks its dominance. At a Hopf point $\alpha = 0$, $\omega^2 = B$, $C = AB$, with $A > 0$ for the remaining root $-A$ to be stable.

## 4 Four-population frequency and the limit of monotonicity

For $\lambda = \alpha + \mathrm{i}\omega$, set $y = \omega^2 > 0$. The real part of $P_4 = 0$ and its imaginary part divided by $\omega$ are, explicitly,

$$\begin{aligned} F(\alpha, y, q) &= \alpha^4 - 6\alpha^2 y + y^2 + c_1(\alpha^3 - 3\alpha y) + c_2(\alpha^2 - y) + c_3\alpha + c_4 = 0, \\ H(\alpha, y, q) &= 4\alpha^3 + 3c_1\alpha^2 + 2c_2\alpha + c_3 - (4\alpha + c_1)y = 0. \end{aligned} \tag{18}$$

These equations determine the nonreal branches; dominance must additionally be checked against the other roots. The required partial derivatives are

$$\begin{aligned} F_\alpha &= 4\alpha^3 - 12\alpha y + 3c_1\alpha^2 - 3c_1 y + 2c_2\alpha + c_3, \\ F_y &= -6\alpha^2 + 2y - 3c_1\alpha - c_2, \\ H_\alpha &= 12\alpha^2 + 6c_1\alpha + 2c_2 - 4y, \qquad & H_y &= -(4\alpha + c_1), \\ F_q &= c_1'(\alpha^3 - 3\alpha y) + c_2'(\alpha^2 - y) + c_3'\alpha + c_4', \\ H_q &= c_1'(3\alpha^2 - y) + 2c_2'\alpha + c_3'. \end{aligned} \tag{19}$$

Implicit differentiation and Cramer's rule give the fully real formula

$$\boxed{\omega' = \frac{H_\alpha F_q - F_\alpha H_q}{2\omega(F_\alpha H_y - F_y H_\alpha)}}. \tag{20}$$

At a simple root, $F_\alpha H_y - F_y H_\alpha = |P_\lambda|^2/(2\omega^2) > 0$: the Jacobian of $(\operatorname{Re} P, \operatorname{Im} P)$ with respect to $(\alpha, \omega)$ has determinant $|P_\lambda|^2$ by the Cauchy–Riemann equations, while $\operatorname{Im} P = \omega H$ and $y = \omega^2$. Thus the numerator in (20) is an exact sign test. At Hopf, $c_3 = c_1 y$, $c_4 = c_2 y - y^2$, and

$$P_4 = (\lambda^2 + y)[\lambda^2 + c_1\lambda + (c_2 - y)]. \tag{21}$$

The noncritical roots are stable precisely when $c_1 > 0$ and $c_2 - y > 0$.

**Proposition 2** (No global nondecreasing positive frequency)**.** *If every retained $\Sigma_{ij} > 0$ and $\tau_i > 0$, then every eigenvalue's imaginary part tends to zero as $q \to \infty$. Hence if $\omega_d(q_*) > 0$, $\omega_d$ cannot be nondecreasing on $[q_*, \infty)$, nor on all subsequent torus wavenumbers. If the $\tau_i$ are distinct, all roots are real for sufficiently large $q$.*

*Proof.* $M(q) = -D^{-1} + E(q)$, where $\|E(q)\| \le C\mathrm{e}^{-\sigma_* q^2/2} \to 0$ and $\sigma_*$ is the smallest retained variance. Continuity of polynomial roots forces the spectrum to the real set $\{-1/\tau_i\}$. A nondecreasing function positive somewhere cannot tend to zero. For distinct $\tau_i$, take disjoint small disks about these real limiting roots. For large $q$, each contains exactly one root counted with multiplicity. Complex conjugation preserves each disk and the spectrum of a real matrix, so its single root must be real. □

This theorem does not prohibit increasing frequency over a finite range.

## 5 What actually increases in the simulations?

The continuous curves interpolate the allowed wavenumbers $q_k = 2\pi k/L$. The following values are ordinary double-precision root calculations; Section 6 proves increase on a smaller interval. At the common onset, (7) gives

$$\omega_3'(q_0) = 0.03316726, \qquad \omega_4'(q_0) = 0.04570080. \tag{22}$$

These derivatives use model-space $q$; divide by ten for physical rad/mm. The constraint $\alpha'(q_0) = 0$ locates the peak of *growth*, not the peak of frequency. There is no reason for $\omega'(q_0)$ to vanish.

| At $p = 0.25$ | Pair becomes dominant | Frequency maximum Wavenumber in rad/mm | Pair loses dominance |
|---|---|---|---|
| Calibrated 3-pop. | 0.634537 | 11.174748 | 17.232247 |
| Calibrated 4-pop. | 1.937968 | No interior maximum | 10.048565 |

The three-population pair rises to $\omega = 0.0967921$ and then decreases, remaining dominant until $\omega = 0.0833387$ at the last switch. The four-population pair rises to a left limit 0.1009536 at its switch; the plotted dominant frequency then drops to zero because a *different, real eigenvalue* becomes rightmost. This is not a smooth frequency maximum, and the complex pair does not disappear there.

| Mode $k$ | Calibrated 3-pop. | Calibrated 4-pop. | Original 4-pop. |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 1 | 0.073439 | 0 | 0.071180 |
| 2 | 0.076059 | 0.074342 | 0.075302 |
| 3 | 0.080000 | 0.080000 | 0.081049 |
| 4 | 0.084551 | 0.086069 | 0.087252 |
| 5 | 0.088966 | 0.091700 | 0.093016 |
| 6 | 0.092675 | 0.096341 | 0.097763 |
| 7 | 0.095304 | 0.099657 | 0.101153 |
| 8 | 0.096640 | 0 | 0.103010 |
| 9 | 0.096593 | 0 | 0 |
| 10 | 0.095167 | 0 | 0 |

Entries are $\omega_d$ in rad/ms. The original four-population matrix is evaluated at its own computed mode-3 onset, $p = 0.2500000483$. Its modes 1–8 do increase; it switches to a real dominant root at 11.158365 rad/mm, before mode 9. Thus the original increasing plot is consistent with the analysis, but does not prove global monotonicity.

Equation (9) also explains the sign quantitatively. At $q_0$, the positive edge contributions sum to 0.11992572 (3-pop.) and 0.15464963 (4-pop.); negative contributions sum to −0.08675846 and −0.10894883. For example, attenuation of E→E contributes +0.08898428 and +0.11141269, whereas E→PV contributes −0.03193637 and −0.04360907. The sums give (22). These are modal sensitivities, not increases in the physical coupling magnitudes.

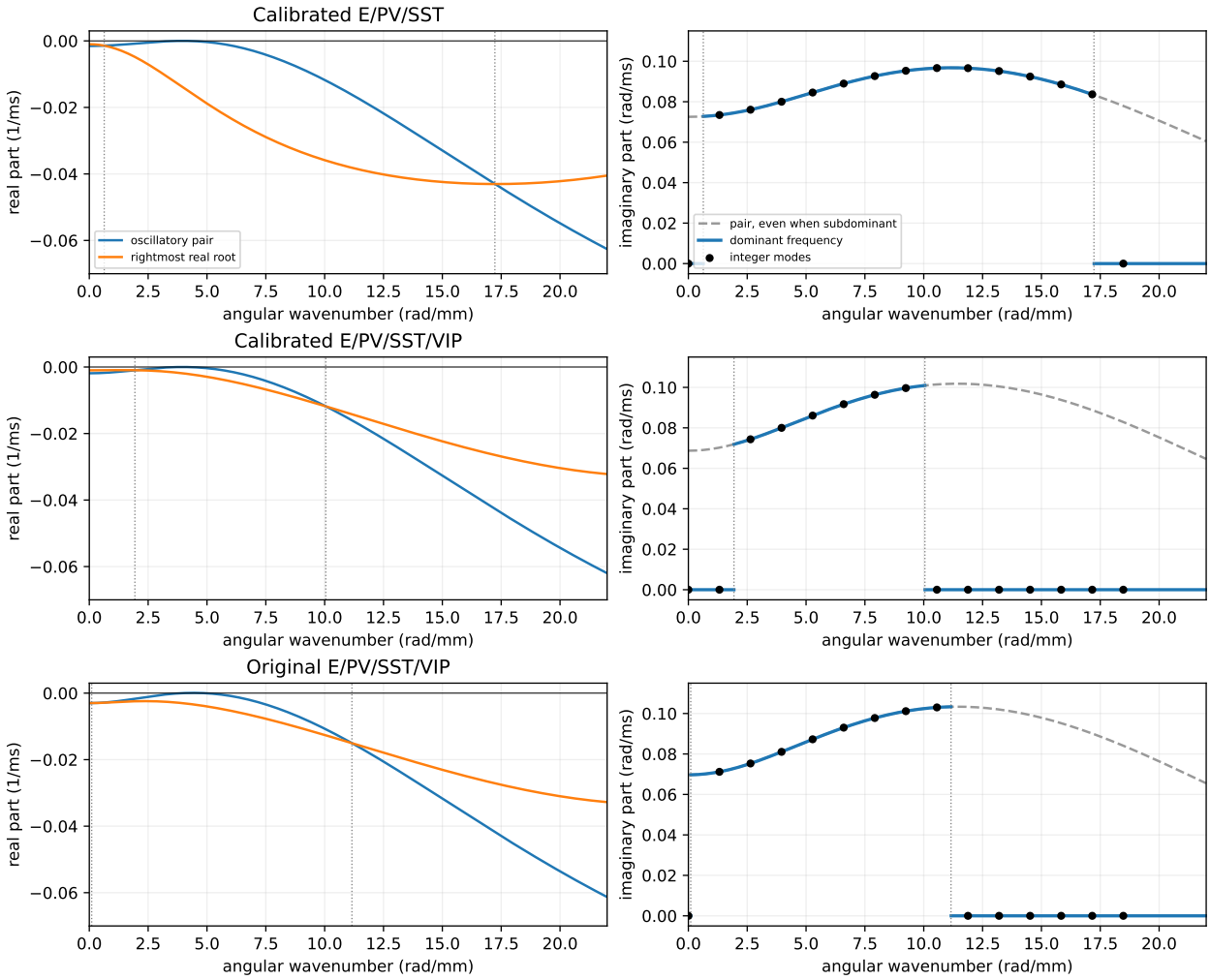



Figure 1: The left panels compare the continued complex pair's real part with the largest real eigenvalue. The right panels distinguish the continued pair's frequency (gray) from the dominant frequency (blue); points mark allowed torus modes. Vertical lines locate dominance switches. The onset is not a frequency maximum. In four populations, the abrupt drop is caused by a real root taking over while the complex pair still oscillates.

The behavior persists in the sampled parameter cases $p = 0.23, 0.2495, 0.25, 0.27$. For three populations, the smooth frequency peak moves from 11.1012 to 11.2445 rad/mm; for four, the upper dominance switch moves from 9.4203 to 10.6089 rad/mm. These sampled continuations are numerical observations, not interval proofs over the entire $(p, q)$ rectangle.

## 6 A certified local increase for both V1 models

The derivative at a single point implies increase on some neighborhood by continuity, provided the root is simple and remains dominant. To give an *explicit* neighborhood without relying on a plotting grid, the supplied interval calculation uses $p = 1/4$ and

$$Q = [0.39597, 0.39599], \qquad Q_{\text{phys}} = [3.9597, 3.9599] \text{ rad/mm}. \tag{23}$$

This interval contains $q_0$. Model constants are interpreted as the exact decimal strings saved in `calibration.json`; the result is conditional on those coefficients, not a certificate about uncertain biological parameters.

Here is the validation argument. Set $\mathcal{F}(x, q) = (\operatorname{Re} P(\alpha + \mathrm{i}\omega, q), \operatorname{Im} P(\alpha + \mathrm{i}\omega, q))$ with $x = (\alpha, \omega)$. Choose a numerical center $x_0$, a rectangle $X = x_0 + [-10^{-4}, 10^{-4}]^2$, and a constant nonsingular approximation $C$ to $\mathcal{F}_x(x_0, q_0)^{-1}$. Interval arithmetic encloses

$$E = I - C\mathcal{F}_x(X, Q), \qquad \mathcal{K} = x_0 - C\mathcal{F}(x_0, Q) + E(X - x_0). \tag{24}$$

For each fixed $q \in Q$, the mean-value theorem gives $x - C\mathcal{F}(x, q) \in \mathcal{K}$ for every $x \in X$. If $\mathcal{K} \subset \operatorname{int} X$ and $\|E\|_\infty < 1$, this map is a contraction of $X$ into itself. Banach's theorem yields exactly one root in $X$ for every $q \in Q$. The same scalar construction encloses every real root. The conjugate rectangle and the disjoint real intervals account for all $N$ polynomial roots. Since their real parts are below $\inf X_\alpha$, the complex pair is dominant throughout $Q$. Invertibility of the Jacobian gives simplicity and a differentiable root branch.

The implementation uses outward-directed 60-digit decimal arithmetic, expands each correctly rounded exponential to its two neighboring decimal values, and evaluates (8) on the entire root rectangle and $Q$. The following are conservative outward roundings of its bounds:

| Certified quantity | 3 populations | 4 populations |
|---|---:|---:|
| Contraction bound $\|E\|_\infty$ | $< 0.00910$ | $< 0.02001$ |
| Lower bound for $\omega'$ | 0.03259 | 0.04407 |
| Upper bound for $\omega'$ | 0.03375 | 0.04739 |
| Other real roots | $[-.013984, -.013783]$ | $[-.039901, -.039700]$ |
| | | $[-.001982, -.001781]$ |

The positive lower bounds prove that *both calibrated models have a strictly increasing dominant frequency on* (23). They do not certify the much wider increasing ranges suggested by Figure 1. The certificate is reproducible with `python -m simulations.v1_interval_certificate`; its full rectangles, Newton images and bounds are saved in `local_monotonicity_certificate.json`.

## 7 The square torus: modes and an exact non-excitation theorem

For $d = 2$, write

$$u(x, y, t) = \sum_{m,n\in\mathbb{Z}} \widehat{u}_{mn}(t)\mathrm{e}^{2\pi\mathrm{i}(mx+ny)/L}, \qquad q_{mn} = \frac{2\pi}{L}(m, n). \tag{25}$$

The linearized forced equation is

$$\partial_t \widehat{u}_{mn} = M(|q_{mn}|, p)\widehat{u}_{mn} + D^{-1}\widehat{I}_{mn}. \tag{26}$$

It does not mix different $(m, n)$, although it mixes populations. A square lattice has discrete rotational/reflection symmetry; radial kernels do not make its allowed directions continuously rotationally invariant. At the selected radius, $m^2 + n^2 = 9$ has exactly the four integer solutions $(\pm 3, 0), (0, \pm 3)$. Thus the $+\mathrm{i}\omega_0$ eigenspace has four complex amplitudes, and the real center space has dimension eight. The computed shell check finds all other shells stable at onset. Degeneracy of the x and y modes does not itself couple them.

**Proposition 3** (An exact axis remains an axis)**.** *Assume the kernels are translation invariant, $f$ is globally Lipschitz, and $I(x, y, t) = I(x, t)$, $u(x, y, 0) = u_0(x)$. Then the unique solution of (1) is independent of $y$ for every $t \geq 0$. In particular, $\widehat{u}_{mn}(t) = 0$ for every $n \neq 0$, even with the full nonlinearity.*

*Proof.* On $C(\mathbb{T}^2_L; \mathbb{R}^N)$ the right-hand side is globally Lipschitz in $u$: convolution is bounded by the kernels' $L^1$ norms and the componentwise sigmoids have Lipschitz constants $g_j/4$. Standard Picard iteration yields uniqueness and global existence for continuous forcing bounded on finite time intervals. For any shift $h$, $v(x, y, t) = u(x, y + h, t)$ solves the same equation, with the same forcing and initial condition. Uniqueness gives $v = u$ for every $h$, proving independence of $y$ and hence the Fourier assertion. □

The statement includes a traveling axial wave $I \propto \cos(qx - \Omega t)$, a static stripe $I \propto \cos(qx)$, and their temporal modulations. Nonlinearity can generate higher x harmonics, but cannot produce transverse modes from exactly zero. Indeed multiplication adds wavevectors,

$$\widehat{u^2_j}_{mn} = \sum_{a,b} \widehat{u}_{j,ab}\widehat{u}_{j,m-a,n-b}; \qquad (a, 0) + (b, 0) = (a + b, 0). \tag{27}$$

The translation proof also covers non-polynomial nonlinearities without requiring convergence of a global Taylor series. More generally Fourier support in an additive lattice subgroup is preserved when the initial data and forcing share that support (equivalently their translation symmetries).

# 8 How transverse modes can grow when they are seeded

Proposition 3 is an invariance theorem, not a stability theorem. Let $U(x,t)$ be an axial solution and put $u = U + v$. To first order,

$$Dv_t = -v + W * [f'(U)v]. \tag{28}$$

If $a_{j,\ell}(t)$ is the x-Fourier coefficient of $f_j'(U_j(x,t))$, then

$$\dot{v}_{i,mn} = -\frac{v_{i,mn}}{\tau_i} + \frac{1}{\tau_i}\sum_j \widehat{W}_{ij}(|q_{mn}|)\sum_{\ell\in\mathbb{Z}} a_{j,m-\ell}(t)v_{j,\ell n}. \tag{29}$$

Different x wavenumbers can now couple within each fixed transverse index $n$. There is still no inhomogeneous source: $v_{\cdot,\cdot n} = 0$ remains zero. For a periodic axial solution, stability of a finite Fourier truncation is determined by its monodromy matrix $\mathcal{U}_n(T,0)$:

$$\gamma_n = T^{-1}\log\rho(\mathcal{U}_n(T,0)). \tag{30}$$

A multiplier of modulus greater than one amplifies some seeded transverse perturbation; multipliers strictly inside the unit circle imply linear decay for that truncation. Equation (26) alone is insufficient around a spatially patterned or time-dependent $U$. For the full function-space problem one needs the corresponding evolution-operator analysis and control of the Fourier truncation; a finite-time trace is not such a proof.

## 8.1 Weak-forcing selection rules, derived by expansion

These rules explain which amplitudes can couple at leading order. They are a formal multiple-scale approximation, separate from the exact theorem above. Set $p = p_0 + \varepsilon^2\mu$, $T = \varepsilon^2 t$, $I = \varepsilon^2 F_2$, and $u = \varepsilon u_1 + \varepsilon^2 u_2 + \cdots$. Let $M(q_0)\Phi = \mathrm{i}\omega_0\Phi$, $\Psi^* M(q_0) = \mathrm{i}\omega_0\Psi^*$, $\Psi^*\Phi = 1$. With $q_x = (q_0, 0)$, $q_y = (0, q_0)$,

$$u_1 = \sum_{\nu\in\{\pm q_x, \pm q_y\}} A_\nu(T)\Phi \mathrm{e}^{\mathrm{i}(\omega_0 t + \nu\cdot x)} + \text{c.c.} \tag{31}$$

Define the quadratic multiplier and nonresonant resolvent by

$$Q_k(v,w) = \tfrac{1}{2}D^{-1}\widehat{W}(k)[f''(0)\odot v\odot w], \qquad R(\Omega,k) = (\mathrm{i}\Omega I_N - M(|k|))^{-1}. \tag{32}$$

A forcing harmonic $F_{\Omega,s}\mathrm{e}^{\mathrm{i}(\Omega t + s\cdot x)}$ contributes $U_{\Omega,s} = R(\Omega,s)D^{-1}F_{\Omega,s}$ to $u_2$ when the inverse exists. At order $\varepsilon^3$, the term $2Q(u_1,u_2)$ is projected by $\Psi^*$ onto each critical harmonic. Matching both frequency and wavevector gives the rules on the next page; the factor two comes from the two quadratic orderings.

## 8.2 Temporal, spatial and traveling pumps

For a uniform temporal pump at $2\omega_0$, the product of $U_{2\omega_0,0}\mathrm{e}^{2\mathrm{i}\omega_0 t}$ and $\overline{A_\mu\Phi}\mathrm{e}^{-\mathrm{i}(\omega_0 t+\mu\cdot x)}$ resonates with mode $\nu$ precisely when $\nu=-\mu$. Hence

$$(\partial_T A_\nu)_{\text{temporal}} = \eta_t\overline{A_{-\nu}}, \qquad \eta_t = 2\Psi^* Q_\nu(\overline{\Phi}, U_{2\omega_0,0}). \tag{33}$$

The same coefficient applies to both axes by symmetry. This is parametric amplification, not an additive source for a previously absent spatial mode.

For a static spatial pump at $s=2q_x$, the product of its second-order response and $A_\mu\Phi\mathrm{e}^{\mathrm{i}(\omega_0 t+\mu\cdot x)}$ resonates when $\nu-\mu=s$. Among the critical modes, only $(\nu,\mu)=(q_x,-q_x)$ satisfies this:

$$(\partial_T A_{q_x})_{\text{spatial}} = \eta_x A_{-q_x}, \qquad \eta_x = 2\Psi^* Q_{q_x}(\Phi, U_{0,2q_x}). \tag{34}$$

The $-2q_x$ harmonic supplies the reverse coupling. For a real cosine the two coefficients coincide, because the static response is real and kernels are even. There is no such leading linear coupling into $A_{\pm q_y}$. An oblique pump $s=q_y-q_x$ could instead couple the two axes; it is not an axial stimulus.

For a traveling pump $(\Omega,s)=(2\omega_0,2q_x)$, resonance with a conjugate critical mode requires $\nu+\mu=2q_x$. Since $|\nu|=|\mu|=q_0$, equality in the triangle inequality forces $\nu=\mu=q_x$. Therefore

$$(\partial_T A_{q_x})_{\text{traveling}} = \eta_w\overline{A_{q_x}}, \qquad \eta_w = 2\Psi^* Q_{q_x}(\overline{\Phi}, U_{2\omega_0,2q_x}). \tag{35}$$

The opposite traveling mode and the transverse pair are not directly resonant with this single traveling harmonic at this order. Reversing propagation exchanges $q_x$ and $-q_x$; the notebook uses $\cos(2q_x\cdot x-2\omega_0 t)$.

Nonlinear amplitude interactions can nevertheless change transverse stability. For example, square symmetry permits terms

$$\partial_T B_\pm = \gamma(T)B_\pm + \kappa(T)\overline{B_\mp} + O(|B|^2), \quad \gamma = a + d(|A_+|^2+|A_-|^2), \quad \kappa = eA_+A_- + \eta_t, \tag{36}$$

where $A_\pm = A_{\pm q_x}$ and $B_\pm = A_{\pm q_y}$; $a,d,e$ are the projected normal-form coefficients. These terms vanish at $B=0$, as the exact theorem requires. If $\gamma,\kappa$ are constant in the chosen rotating frame, the matrix acting on $(B_+,\overline{B}_-)$ and its exponents are

$$\begin{pmatrix}\gamma & \kappa\\ \overline{\kappa} & \overline{\gamma}\end{pmatrix}, \qquad \Lambda_\pm = \operatorname{Re}\gamma \pm \sqrt{|\kappa|^2-(\operatorname{Im}\gamma)^2}. \tag{37}$$

Positive real part permits seeded growth; zero initial $B$ still stays zero. For varying coefficients one must instead solve the variational equation.

# 9 Direct simulations, frequency interpretation and reproducibility

The notebook integrates (1) using Fourier convolution and RK4, with the coefficients in Report 1. The 1D runs use 64 points, a 1 ms step, 80 s duration and saved output every 10 ms. At $p = 0.2495$, homogeneous temporal forcing at $2\omega_0$ or static spatial forcing at mode 6 each produces an oscillatory mode-3 response above the decaying unforced control. The table averages $|\Psi^* \widehat{u}_{k_0}|$ over the final 16 s:

| Late projected mode-3 magnitude | 3 populations | 4 populations |
|---|---|---|
| Unforced | $2.58 \times 10^{-5}$ | $1.47 \times 10^{-5}$ |
| Temporal only | $1.84 \times 10^{-3}$ | $1.37 \times 10^{-3}$ |
| Static spatial only | $2.23 \times 10^{-3}$ | $1.74 \times 10^{-3}$ |

The resolved late response is about 12.75 Hz, near $1000\omega_0/(2\pi) = 12.7324$ Hz. Temporal forcing is at 25.4648 Hz, so this is the predicted near-subharmonic response. Static forcing has zero temporal frequency; its oscillatory response is generated by the field dynamics. The late FFT resolution is 0.0625 Hz (16 s window). The dispersion curve $\omega_d(q, p)$ describes the unforced linearization around zero. It is not the forced oscillation frequency for every nonlinear state. Damped eigenfrequencies need not appear as persistent peaks in a forced or noisy trace, and branch switching does not require a visible nonlinear oscillation to stop abruptly.

The 2D runs use a $32 \times 32$ square torus, 1 ms steps and 60 s duration. The reported transverse norm is the RMS over populations and grid points of $u - \langle u \rangle_y$. Exactly axial initial data remain axial. With a nonzero transverse seed, the outcomes are:

| Transverse RMS | 3 populations | 4 populations |
|---|---|---|
| Initial seeded value | $1.225 \times 10^{-4}$ | $1.061 \times 10^{-4}$ |
| Final, axial traveling drive | $1.327 \times 10^{-6}$ | $3.888 \times 10^{-8}$ |
| Final, axial static drive | $9.949 \times 10^{-7}$ | $7.064 \times 10^{-7}$ |
| Final, uniform temporal drive | $2.298 \times 10^{-3}$ | $1.009 \times 10^{-3}$ |
| Final, unforced | $1.522 \times 10^{-5}$ | $6.859 \times 10^{-6}$ |

These examples demonstrate decay under the tested axial drives and growth under uniform temporal pumping. They neither exclude transverse instability under other axial drives nor estimate a converged Floquet exponent. The exact answer is: an axis wave cannot create transverse modes from perfectly axial data; it may change the growth of transverse modes already present.

Validation checks preserve the connectivity masks, compare sparse determinants and eigenvalue derivatives, verify critical square-lattice shells and the Gaussian high-wavenumber stability tail, and compare time steps. A 32 versus 48 grid comparison at 2 s for the 2D spatial case has relative errors $1.41 \times 10^{-5}$ and $3.42 \times 10^{-5}$; this short check is not an error bound for the entire 60 s trajectory.

**Reproduce.** The executed notebook is `simulations/v1_three_four_population.ipynb`. The full coefficient and simulation records are in `simulations/output/v1_extension/`. `v1_dispersion_detail.py` generates the branch figure and switch data; `v1_interval_certificate.py` validates the local increase. The LaTeX sources and PDF builder are in `notes/`. Report 1 documents the biological references; all spectral and symmetry results here are derived from the explicitly stated field equation.